

EXPERIMENT-10: Express REST API CRUD application.

Develop an Express web application that can interact with REST APIs to perform CRUD operations on student data using Postman.

Objective : Build a RESTful student API with create/read/update/delete endpoints.

Tools : Node.js, Express, REST, Postman

Outcome : A student management REST API that supports full CRUD via HTTP requests.

Answer:

Define Express routes for student CRUD operations and use JSON body parsing to handle request data.

Objective

Create a RESTful student management API using Node.js and Express. Test the API with Postman using CRUD operations:

- ****Create**** a student
- ****Read**** all students or one student
- ****Update**** a student
- ****Delete**** a student

Requirements

- Node.js installed
- Postman installed, or any REST client

No database is required. Student data is kept in memory and resets whenever the server restarts.

1. Open the project

Open PowerShell and run:

```
```powershell
cd "D:\java full stack\NODE JS\week-10"
```
```

2. Install Express

Run this once:

```
```powershell
npm install
```
```

3. Start the server

```
```powershell
npm start
```
```

You should see:

```
```text
Student API running at http://localhost:3000
```
```

Keep this terminal open. Open this URL in a browser to check the API:

```
```text
http://localhost:3000/
```
```

Stop the server with `Ctrl + C`.

4. Test with Postman

Create a Postman request for each example below. For POST and PUT requests, select ****Body > raw > JSON****.

Read all students

- Method: `GET`
- URL: `http://localhost:3000/api/students`

Read one student

- Method: `GET`
- URL: `http://localhost:3000/api/students/1`

Create a student

- Method: `POST`
- URL: `http://localhost:3000/api/students`
- Body:

```
```json
{
 "name": "Kiran",
 "email": "kiran@example.com",
 "course": "Express"
}
```
```

Expected status: `201 Created`.

Update a student

- Method: `PUT`
- URL: `http://localhost:3000/api/students/1`
- Body:

```
```json
{
 "name": "Anita Sharma",
 "email": "anita.sharma@example.com",
 "course": "Express REST API"
}
```
```

Expected status: `200 OK`.

Delete a student

- Method: `DELETE`
- URL: `http://localhost:3000/api/students/2`

Expected status: `200 OK`.

API route summary

| Method | URL | Purpose |
|--------|-------------------|-------------------|
| GET | /api/students | Read all students |
| GET | /api/students/:id | Read one student |
| POST | /api/students | Create a student |
| PUT | /api/students/:id | Update a student |
| DELETE | /api/students/:id | Delete a student |

How the code works

- `express()` creates the application.
- `express.json()` reads JSON sent by Postman.
- `app.get()` defines read routes.
- `app.post()` defines the create route.
- `app.put()` defines the update route.
- `app.delete()` defines the delete route.
- HTTP status `201` means a resource was created.
- HTTP status `400` means the request data is invalid.
- HTTP status `404` means the student or route was not found.

Common errors

`EADDRINUSE: port 3000`

Another server is using port 3000. Either stop it with `Ctrl + C`, or run:

```
```powershell
```

```
$env:PORT=4000
```

```
npm start
```

```
...
```

Then use `http://localhost:4000` in Postman.

```
`Cannot find module 'express'`
```

Run:

```
"powershell"
```

```
npm install
```

**File Name: index.html**

**code:**

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8" />
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
```

```
<title>Student CRUD Demo</title>
```

```
<style>
```

```
body {
```

```
 font-family: Arial, sans-serif;
```

```
 margin: 40px;
```

```
 background: #f5f7fb;
```

```
 color: #222;
```

```
}
```

```
.box {
```

```
 background: #fff;
```

```
 border-radius: 12px;
```

```
 padding: 20px;
```

```
 box-shadow: 0 3px 10px rgba(0,0,0,0.08);
```

```
 margin-bottom: 20px;
```

```
 max-width: 900px;
```

```
}
```

```
input, button, textarea {
```

```
width: 100%;
padding: 10px;
margin: 8px 0;
box-sizing: border-box;
font-size: 14px;
}
button {
 background: #1976d2;
 border: none;
 color: white;
 border-radius: 6px;
 cursor: pointer;
}
button.delete {
 background: #d32f2f;
}
textarea {
 min-height: 180px;
 font-family: Consolas, monospace;
}
.row {
 display: flex;
 gap: 20px;
 flex-wrap: wrap;
}
.col {
 flex: 1;
 min-width: 280px;
}
</style>
</head>
<body>
<h1>Student CRUD REST API Demo</h1>

<div class="box">
 <h2>Create Student</h2>
```

```
<div class="row">
 <div class="col">
 <input id="name" type="text" placeholder="Name" value="Kiran" />
 <input id="email" type="text" placeholder="Email" value="kiran@example.com" />
 <input id="course" type="text" placeholder="Course" value="Express" />
 <button id="createBtn">Create Student</button>
 </div>
 <div class="col">
 <textarea id="createResult" placeholder="Create response will appear here..."></textarea>
 </div>
</div>
</div>
```

```
<div class="box">
 <h2>Read Students</h2>
 <button id="readBtn">Get All Students</button>
 <textarea id="readResult" placeholder="Fetch response will appear here..."></textarea>
</div>
```

```
<div class="box">
 <h2>Update Student</h2>
 <input id="updateId" type="number" placeholder="Student ID" value="1" />
 <input id="updateName" type="text" placeholder="Updated Name" value="Anita Sharma" />
 <input id="updateEmail" type="text" placeholder="Updated Email"
value="anita.sharma@example.com" />
 <input id="updateCourse" type="text" placeholder="Updated Course" value="Express REST API" />
 <button id="updateBtn">Update Student</button>
 <textarea id="updateResult" placeholder="Update response will appear here..."></textarea>
</div>
```

```
<div class="box">
 <h2>Delete Student</h2>
 <input id="deleteId" type="number" placeholder="Student ID" value="2" />
 <button id="deleteBtn" class="delete">Delete Student</button>
 <textarea id="deleteResult" placeholder="Delete response will appear here..."></textarea>
</div>
```

```
<script>
```

```
const apiBase = 'http://localhost:3000';
```

```
async function request(url, options = {}) {
 const response = await fetch(url, options);
 const data = await response.json();
 return { response, data };
}
```

```
document.getElementById('createBtn').addEventListener('click', async () => {
 const result = document.getElementById('createResult');
 const payload = {
 name: document.getElementById('name').value,
 email: document.getElementById('email').value,
 course: document.getElementById('course').value
 };
 try {
 const { response, data } = await request(`${apiBase}/api/students`, {
 method: 'POST',
 headers: { 'Content-Type': 'application/json' },
 body: JSON.stringify(payload)
 });
 result.value = JSON.stringify(response.ok ? data : { error: data.error }, null, 2);
 } catch (error) {
 result.value = `Error: ${error.message}`;
 }
});
```

```
 try {
 const { response, data } = await request(`${apiBase}/api/students`, {
 method: 'POST',
 headers: { 'Content-Type': 'application/json' },
 body: JSON.stringify(payload)
 });
 result.value = JSON.stringify(response.ok ? data : { error: data.error }, null, 2);
 } catch (error) {
 result.value = `Error: ${error.message}`;
 }
});
```

```
document.getElementById('readBtn').addEventListener('click', async () => {
 const result = document.getElementById('readResult');
 try {
 const { response, data } = await request(`${apiBase}/api/students`);
 result.value = JSON.stringify(response.ok ? data : { error: data.error }, null, 2);
 }
});
```



```
 } catch (error) {
 result.value = `Error: ${error.message}`;
 }
 });
```

```
document.getElementById('updateBtn').addEventListener('click', async () => {
 const result = document.getElementById('updateResult');
 const id = document.getElementById('updateId').value;
 const payload = {
 name: document.getElementById('updateName').value,
 email: document.getElementById('updateEmail').value,
 course: document.getElementById('updateCourse').value
 };

```

```
 try {
 const { response, data } = await request(`${apiBase}/api/students/${id}`, {
 method: 'PUT',
 headers: { 'Content-Type': 'application/json' },
 body: JSON.stringify(payload)
 });

```

```
 result.value = JSON.stringify(response.ok ? data : { error: data.error }, null, 2);
 } catch (error) {
 result.value = `Error: ${error.message}`;
 }
});
```

```
document.getElementById('deleteBtn').addEventListener('click', async () => {
 const result = document.getElementById('deleteResult');
 const id = document.getElementById('deleteId').value;
```

```
 try {
 const { response, data } = await request(`${apiBase}/api/students/${id}`, {
 method: 'DELETE'
 });

```

```

 result.value = JSON.stringify(response.ok ? data : { error: data.error }, null, 2);
 } catch (error) {
 result.value = `Error: ${error.message}`;
 }
});
</script>
</body>
</html>

```

**File Name : package-lock.json**

**Code:**

```

{
 "name": "student-crud-api",
 "version": "1.0.0",
 "lockfileVersion": 3,
 "requires": true,
 "packages": {
 "": {
 "name": "student-crud-api",
 "version": "1.0.0",
 "dependencies": {
 "express": "^4.21.2"
 }
 },
 "node_modules/accepts": {
 "version": "1.3.8",
 "resolved": "https://registry.npmjs.org/accepts/-/accepts-1.3.8.tgz",
 "integrity": "sha512-5e8e281C1e086c39bA87sRP+J0cBnzQdE8JPeE6VJb96GwbcYqX7YAodWjZL31nQNVFndGtO87LbeK+Uw==",
 "license": "MIT",
 "dependencies": {
 "mime-types": "~2.1.34",
 "negotiator": "0.6.3"
 }
 },
 "engines": {

```

```
 "node": ">= 0.6"
 }
},
"node_modules/array-flatten": {
 "version": "1.1.1",
 "resolved": "https://registry.npmjs.org/array-flatten/-/array-flatten-1.1.1.tgz",
 "integrity": "sha512-PCVAQswWemu6UdxsDFFX/+gVeYqKAod3D3UVm91jHwynguOwAvYPhx8nNIM++NqRcK6CxxpUafjmh
 IdKiHibqg==",
 "license": "MIT"
},
"node_modules/body-parser": {
 "version": "1.20.6",
 "resolved": "https://registry.npmjs.org/body-parser/-/body-parser-1.20.6.tgz",
 "integrity": "sha512-PCVAQswWemu6UdxsDFFX/+gVeYqKAod3D3UVm91jHwynguOwAvYPhx8nNIM++NqRcK6CxxpUafjmh
 p5tAzS57i5MV9fZFDj9LeliTZEufbSe2eDozP+ElheSUq1m74CRq1jl4mYNDdVs9vQztXFLuk/Gd6BWTdwr
 J5g==",
 "license": "MIT",
 "dependencies": {
 "bytes": "~3.1.2",
 "content-type": "~1.0.5",
 "debug": "2.6.9",
 "depd": "2.0.0",
 "destroy": "~1.2.0",
 "http-errors": "~2.0.1",
 "iconv-lite": "~0.4.24",
 "on-finished": "~2.4.1",
 "qs": "~6.15.1",
 "raw-body": "~2.5.3",
 "type-is": "~1.6.18",
 "unpipe": "~1.0.0"
 },
 "engines": {
 "node": ">= 0.8",
 "npm": "1.2.8000 || >= 1.4.16"
 }
},
```

```
"node_modules/bytes": {
 "version": "3.1.2",
 "resolved": "https://registry.npmjs.org/bytes/-/bytes-3.1.2.tgz",
 "integrity": "sha512-1Fb593519t2e49u3v21bE8L577E4Jp32o8EoXjU/7Np2AoSRkDzF28B8YZjUv2jgMjr0kZxHVHw+Ht8Q==",
 "license": "MIT",
 "engines": {
 "node": ">= 0.8"
 }
},
"node_modules/call-bind-apply-helpers": {
 "version": "1.0.2",
 "resolved": "https://registry.npmjs.org/call-bind-apply-helpers/-/call-bind-apply-helpers-1.0.2.tgz",
 "integrity": "sha512-+v5W0NnDq0xV6Fp5MbnXp6UWMQ/4V1mHw40kOqjL19L7bN73ZTK1dv/DzY12Omuft7Q1iS/dm1xh0HQA==",
 "license": "MIT",
 "dependencies": {
 "es-errors": "^1.3.0",
 "function-bind": "^1.1.2"
 },
 "engines": {
 "node": ">= 0.4"
 }
},
"node_modules/call-bound": {
 "version": "1.0.4",
 "resolved": "https://registry.npmjs.org/call-bound/-/call-bound-1.0.4.tgz",
 "integrity": "sha512-+ys997U96po4Kx/ABpBCqhA9EuxJaQWDQg7295H4hBphv3lZg0boBKuwYpt4YXp6MZ5AmZQnU/tyMTlRpaSejg==",
 "license": "MIT",
 "dependencies": {
 "call-bind-apply-helpers": "^1.0.2",
 "get-intrinsic": "^1.3.0"
 }
}
```

```

},
"engines": {
 "node": ">= 0.4"
},
"funding": {
 "url": "https://github.com/sponsors/ljharb"
}
},
"node_modules/content-disposition": {
 "version": "0.5.4",
 "resolved": "https://registry.npmjs.org/content-disposition/-/content-disposition-0.5.4.tgz",
 "integrity": "sha512-FveZTNuGw04cxIAiWbzi6zTAL/IhehaWbTtgluJh4/E95DqMwTmha3KZN1aAWA8cFIhHzMZUvLevw5Rqk+tSQ==",
 "license": "MIT",
 "dependencies": {
 "safe-buffer": "5.2.1"
 },
 "engines": {
 "node": ">= 0.6"
 }
},
"node_modules/content-type": {
 "version": "1.0.5",
 "resolved": "https://registry.npmjs.org/content-type/-/content-type-1.0.5.tgz",
 "integrity": "sha512-nTjqfcBFEipKdXCv4YDQWCfmcLZKm81ldF0pAopTvyrFGVbcR6P/VAAd5G7N+0tTr8QqiU0tFadD6FK4NtJwOA==",
 "license": "MIT",
 "engines": {
 "node": ">= 0.6"
 }
},
"node_modules/cookie": {
 "version": "0.7.2",
 "resolved": "https://registry.npmjs.org/cookie/-/cookie-0.7.2.tgz",

```

```
 "integrity": "sha512-
yki5XnKuf750l50uGTlt6kKILY4nQ1eNIQatoXEBYz5dWgnKqbnqmTrBE5B4N7lrMJKQ2ytWMiTO2o0v6E
w/w==",
 "license": "MIT",
 "engines": {
 "node": ">= 0.6"
 }
 },
 "node_modules/cookie-signature": {
 "version": "1.0.7",
 "resolved": "https://registry.npmjs.org/cookie-signature/-/cookie-signature-1.0.7.tgz",
 "integrity": "sha512-
NXdYc3dLr47pBkpUCHtKSwiOQXLVn8dZEuywboCOJY/osA0wFSLISawr3KN8qXJEyX66FcONTH8EIlVuK0
yyFA==",
 "license": "MIT"
 },
 "node_modules/debug": {
 "version": "2.6.9",
 "resolved": "https://registry.npmjs.org/debug/-/debug-2.6.9.tgz",
 "integrity": "sha512-
bC7ElrdJaJnPbAP+1EotYvqZsb3ecl5wi6Bfi6BJTUcNowp6cvspg0jXznRTKDjm/E7AdgFBVeAPVMNcKGsH
MA==",
 "license": "MIT",
 "dependencies": {
 "ms": "2.0.0"
 }
 },
 "node_modules/depd": {
 "version": "2.0.0",
 "resolved": "https://registry.npmjs.org/depd/-/depd-2.0.0.tgz",
 "integrity": "sha512-
g7nH6P6dyDioJogAAGprGpCtVImJhpPk/roCzdb3flh61/s/nPsfR6onyMwkCAR/OlC3yBC0lESvUoQEAssl
rw==",
 "license": "MIT",
 "engines": {
 "node": ">= 0.8"
 }
 }
```

```

},
"node_modules/destroy": {
 "version": "1.2.0",
 "resolved": "https://registry.npmjs.org/destroy/-/destroy-1.2.0.tgz",
 "integrity": "sha512-2sJGJTaXIIaR1w4iJSNoN0hnMY7Gpc/n8D4qSCJw8QqFWXf7cuAgnEHxBpweaVcPevC2l3KpjYCx3NypQ
QgaJg==",
 "license": "MIT",
 "engines": {
 "node": ">= 0.8",
 "npm": "1.2.8000 || >= 1.4.16"
 }
},
"node_modules/dunder-proto": {
 "version": "1.0.1",
 "resolved": "https://registry.npmjs.org/dunder-proto/-/dunder-proto-1.0.1.tgz",
 "integrity": "sha512-KIN/nDJBQRcXw0MLVhZE9iQHmG68qAVIBg9CqmUYjmQlhgij9U5MFvrqkUL5FbtyyzZuOeOt0zdeRe4U
Y7ct+A==",
 "license": "MIT",
 "dependencies": {
 "call-bind-apply-helpers": "^1.0.1",
 "es-errors": "^1.3.0",
 "gopd": "^1.2.0"
 },
 "engines": {
 "node": ">= 0.4"
 }
},
"node_modules/ee-first": {
 "version": "1.1.1",
 "resolved": "https://registry.npmjs.org/ee-first/-/ee-first-1.1.1.tgz",
 "integrity": "sha512-WMwm9LhRUo+WUaRN+vRuETqG89IgZphVSNkdFgeb6sS/E4OrDIN7t48CAewSHXc6C8lefD8KKfr5vY6
1brQlow==",
 "license": "MIT"
},

```

```
"node_modules/encodeurl": {
 "version": "2.0.0",
 "resolved": "https://registry.npmjs.org/encodeurl/-/encodeurl-2.0.0.tgz",
 "integrity": "sha512-Q0n9HRi4m6JuGIV1eFlmvJB7ZEVxu93lrMyiMsGC0lrMJMWzRgx6WGquyfQgZVb31vhGgXnfmPNNXmxnOkRBrg==",
 "license": "MIT",
 "engines": {
 "node": ">= 0.8"
 }
},
"node_modules/es-define-property": {
 "version": "1.0.1",
 "resolved": "https://registry.npmjs.org/es-define-property/-/es-define-property-1.0.1.tgz",
 "integrity": "sha512-e3nRfgfUZ4rNGL232gUgX06QNYyez04KdjFrF+LTRoOXmrOgFKDg4BCdsjW8EnT69eqdYGmRpJwiPVYNrCaW3g==",
 "license": "MIT",
 "engines": {
 "node": ">= 0.4"
 }
},
"node_modules/es-errors": {
 "version": "1.3.0",
 "resolved": "https://registry.npmjs.org/es-errors/-/es-errors-1.3.0.tgz",
 "integrity": "sha512-e3nRfgfUZ4rNGL232gUgX06QNYyez04KdjFrF+LTRoOXmrOgFKDg4BCdsjW8EnT69eqdYGmRpJwiPVYNrCaW3g==",
 "license": "MIT",
 "engines": {
 "node": ">= 0.4"
 }
},
"node_modules/es-object-atoms": {
```



```
 "integrity": "sha512-
HWcBoN6NileqtSydK2FqHbS/LoDd2pqrnQHLYJzBj4kOp/ky2MWMN694xOfkK8/SnUsW2DH7EfyVlydK
Csm1Zw==",
 "license": "MIT",
 "dependencies": {
 "es-errors": "^1.3.0"
 },
 "engines": {
 "node": ">= 0.4"
 }
 },
 "node_modules/escape-html": {
 "version": "1.0.3",
 "resolved": "https://registry.npmjs.org/escape-html/-/escape-html-1.0.3.tgz",
 "integrity": "sha512-
NiSupZ4OeuGwr68IGleyM/ksIZMJodUGOSZ/FSnTxcrekbvqrgdUxIJOMpijaKZVjAJrWrGs/6Jy8OMuyj9
ow==",
 "license": "MIT"
 },
 "node_modules/etag": {
 "version": "1.8.1",
 "resolved": "https://registry.npmjs.org/etag/-/etag-1.8.1.tgz",
 "integrity": "sha512-
alL5Fx7mawVa300al2BnEE4iNvo1qETxLrPI/o05L7z6go7fCw1J6EQmbK4FmJ2AS7kgVF/KEZWufBfdCIM
cPg==",
 "license": "MIT",
 "engines": {
 "node": ">= 0.6"
 }
 },
 "node_modules/express": {
 "version": "4.22.2",
 "resolved": "https://registry.npmjs.org/express/-/express-4.22.2.tgz",
 "integrity": "sha512-
luL+Elrou2ZvCFHs18/Clzy2Nzvo25nZ1/D2elZlz7c+QUayAcYoiM2BthCjs+EBHVpjYjcuLDAiCWgelX3X1Q
==",
 "license": "MIT",
```

```
"dependencies": {
 "accepts": "~1.3.8",
 "array-flatten": "1.1.1",
 "body-parser": "~1.20.5",
 "content-disposition": "~0.5.4",
 "content-type": "~1.0.4",
 "cookie": "~0.7.1",
 "cookie-signature": "~1.0.6",
 "debug": "2.6.9",
 "depd": "2.0.0",
 "encodeurl": "~2.0.0",
 "escape-html": "~1.0.3",
 "etag": "~1.8.1",
 "finalhandler": "~1.3.1",
 "fresh": "~0.5.2",
 "http-errors": "~2.0.0",
 "merge-descriptors": "1.0.3",
 "methods": "~1.1.2",
 "on-finished": "~2.4.1",
 "parseurl": "~1.3.3",
 "path-to-regexp": "~0.1.12",
 "proxy-addr": "~2.0.7",
 "qs": "~6.15.1",
 "range-parser": "~1.2.1",
 "safe-buffer": "5.2.1",
 "send": "~0.19.0",
 "serve-static": "~1.16.2",
 "setprototypeof": "1.2.0",
 "statuses": "~2.0.1",
 "type-is": "~1.6.18",
 "utils-merge": "1.0.1",
 "vary": "~1.1.2"
},
"engines": {
 "node": ">= 0.10.0"
},
```

```
"funding": {
 "type": "opencollective",
 "url": "https://opencollective.com/express"
},
"node_modules/finalhandler": {
 "version": "1.3.2",
 "resolved": "https://registry.npmjs.org/finalhandler/-/finalhandler-1.3.2.tgz",
 "integrity": "sha512-aA4RyPcd3badbdABGDuTXCMTtOneUCAYH/gxoYRTZIIJdFOYPWuGqiAslrhNnnqdXGswYk6dGujem4w80UJFhg==",
 "license": "MIT",
 "dependencies": {
 "debug": "2.6.9",
 "encodeurl": "~2.0.0",
 "escape-html": "~1.0.3",
 "on-finished": "~2.4.1",
 "parseurl": "~1.3.3",
 "statuses": "~2.0.2",
 "unpipe": "~1.0.0"
 },
 "engines": {
 "node": ">= 0.8"
 }
},
"node_modules/forwarded": {
 "version": "0.2.0",
 "resolved": "https://registry.npmjs.org/forwarded/-/forwarded-0.2.0.tgz",
 "integrity": "sha512-buRG0fpBtRHSTCOASE6hD258tEubFoRLb4ZNA6NxMVHNw2gOcwHo9wyablzMzOA5z9xA9L1KNjk/Nt6MT9aYow==",
 "license": "MIT",
 "engines": {
 "node": ">= 0.6"
 }
},
"node_modules/fresh": {
```

```
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 "es-errors": "^1.3.0",
 "es-object-atoms": "^1.1.1",
 "function-bind": "^1.1.2",
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 "gopd": "^1.2.0",
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 "license": "MIT",

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 "setprototypeof": "~1.2.0",
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 "license": "ISC"
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 "ipaddr.js": "1.9.1"
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 "http-errors": "~2.0.1",
 "iconv-lite": "~0.4.24",
 "unpipe": "~1.0.0"
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 "etag": "~1.8.1",
 "fresh": "~0.5.2",
 "http-errors": "~2.0.1",
 "mime": "1.6.0",
 "ms": "2.1.3",
 "on-finished": "~2.4.1",
 "range-parser": "~1.2.1",
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 "license": "MIT",
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 "object-inspect": "^1.13.4",
 "side-channel-list": "^1.0.1",
 "side-channel-map": "^1.0.1",
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 "funding": {
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 "url": "https://github.com/sponsors/ljharb"
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 "license": "MIT",
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 "get-intrinsic": "^1.2.5",
 "object-inspect": "^1.13.3"
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 "object-inspect": "^1.13.3",
 "side-channel-map": "^1.0.1"
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 "url": "https://github.com/sponsors/ljharb"
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 "mime-types": "~2.1.24"
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"node_modules/unpipe": {
 "version": "1.0.0",
 "resolved": "https://registry.npmjs.org/unpipe/-/unpipe-1.0.0.tgz",
 "integrity": "sha512-pjy2bYhSsufwWIKwPc+l3cN7+wuJlK6uz0YdJEOlQDbl6jo/YIPi4mb8agUkVC8BF7V8NuzeyPNqRksA3hzt
KQ==",
 "license": "MIT",
 "engines": {
 "node": ">= 0.8"
 }
},
```

```

"node_modules/utils-merge": {
 "version": "1.0.1",
 "resolved": "https://registry.npmjs.org/utils-merge/-/utils-merge-1.0.1.tgz",
 "integrity": "sha512-lpE64qFzpt6pbqIvO8rXklWGeZJpVt8zkiB7B16u7b+vGQgS64YU6g2DfLZ5n4b3EgWys3Eo6tqYKqH5A==",
 "license": "MIT",
 "engines": {
 "node": ">= 0.4.0"
 }
},
"node_modules/vary": {
 "version": "1.1.2",
 "resolved": "https://registry.npmjs.org/vary/-/vary-1.1.2.tgz",
 "integrity": "sha512-BNGbWVld0eUPabkhXUVm0j8uuvREyTh5ovRa/dyow/BqAbZJyC+5fU+IzQOzmAKzYqYRAISoRhdQr3eIZ/PXqg==",
 "license": "MIT",
 "engines": {
 "node": ">= 0.8"
 }
}
}

```

**File Name :** package.json

Code:

```

{
 "name": "student-crud-api",
 "version": "1.0.0",
 "description": "Experiment 10: Express REST API CRUD application",
 "main": "server.js",
 "scripts": {
 "start": "node server.js",
 "dev": "node --watch server.js"
 },
 "dependencies": {

```

```
"express": "^4.21.2"
}
}
```

**File Name :** server.js

**Code:**

```
const express = require("express");

const app = express();
const PORT = process.env.PORT || 3000;

// Allows Express to read JSON request bodies.
app.use(express.json());
app.use(express.static(__dirname));

let nextId = 4;
let students = [
 { id: 1, name: "Anita", email: "anita@example.com", course: "Java" },
 { id: 2, name: "Rahul", email: "rahul@example.com", course: "Node.js" },
 { id: 3, name: "Meena", email: "meena@example.com", course: "Database" }
];

function findStudent(id) {
 return students.find((student) => student.id === Number(id));
}

app.get("/", (req, res) => {
 res.json({
 message: "Student CRUD REST API",
 endpoints: {
 getAll: "GET /api/students",
 getOne: "GET /api/students/:id",
 create: "POST /api/students",
 update: "PUT /api/students/:id",
 delete: "DELETE /api/students/:id"
 }
 })
})
```

```

});

});

// READ: return all students.
app.get("/api/students", (req, res) => {
 res.status(200).json(students);
});

// READ: return one student by ID.
app.get("/api/students/:id", (req, res) => {
 const student = findStudent(req.params.id);
 if (!student) {
 return res.status(404).json({ error: "Student not found" });
 }
 res.status(200).json(student);
});

// CREATE: add a student from the JSON request body.
app.post("/api/students", (req, res) => {
 const { name, email, course } = req.body;

 if (!name || !email || !course) {
 return res.status(400).json({
 error: "name, email, and course are required"
 });
 }

 const student = {
 id: nextId++,
 name: String(name).trim(),
 email: String(email).trim(),
 course: String(course).trim()
 };

 students.push(student);
 res.status(201).json(student);

```

```
});
```

```
// UPDATE: replace the editable fields of one student.
```

```
app.put("/api/students/:id", (req, res) => {
 const student = findStudent(req.params.id);
 if (!student) {
 return res.status(404).json({ error: "Student not found" });
 }

 const { name, email, course } = req.body;
 if (!name || !email || !course) {
 return res.status(400).json({
 error: "name, email, and course are required"
 });
 }

 student.name = String(name).trim();
 student.email = String(email).trim();
 student.course = String(course).trim();
 res.status(200).json(student);
});
```

```
// DELETE: remove one student by ID.
```

```
app.delete("/api/students/:id", (req, res) => {
 const studentIndex = students.findIndex(
 (student) => student.id === Number(req.params.id)
);

 if (studentIndex === -1) {
 return res.status(404).json({ error: "Student not found" });
 }

 const deletedStudent = students.splice(studentIndex, 1)[0];
 res.status(200).json({
 message: "Student deleted successfully",
 student: deletedStudent
 });
});
```

```
});
});
```

```
app.use((req, res) => {
 res.status(404).json({ error: "Route not found" });
});
```

```
app.listen(PORT, () => {
 console.log(`Student API running at http://localhost:${PORT}`);
});
```